WEEK 4 · LESSON 19 OF 24 · HANDOUT

Bayes' Theorem: Updating Beliefs with Evidence

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=HZGCoVF3YvM — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 2: Uncertainty (probability and Bayes) — https://www.youtube.com/watch?v=D8RRq3TbtHU

Your goal in this lesson: Use Bayes' theorem to update a probability when new evidence arrives, avoid the base-rate mistake, and see how a spam filter uses the same idea.

A surprising test result

A disease affects **1%** of people. A test catches **95%** of sick people, but also wrongly says "positive" for **10%** of healthy people. Your test is positive. How likely is it that you are sick?

Many people answer "95%". The real answer is **under 9%**. Bayes' theorem explains why.

Count it with 10,000 people

	Test positive	Test negative	Total
Sick (1%)	95	5	100
Healthy (99%)	990	8,910	9,900
Total	1,085	8,915	10,000

Of the 1,085 people who test positive, only 95 are sick: 95 ÷ 1,085 ≈ **8.8%**. The disease is so rare that false alarms from the large healthy group outnumber the true cases.

Bayes' theorem

> $P(A \mid B) = P(B \mid A) \times P(A) \div P(B)$

In words: **new belief = how well the evidence fits × old belief ÷ how common the evidence is.**

- **Prior** P(sick) = 0.01 — what we believed before the test.
- **Likelihood** P(positive | sick) = 0.95 — how well the evidence fits.
- **Evidence** P(positive) = 0.01 × 0.95 + 0.99 × 0.10 = 0.1085.
- **Posterior** P(sick | positive) = 0.95 × 0.01 ÷ 0.1085 ≈ 0.088.

```
def bayes(prior, p_evidence_if_true, p_evidence_if_false):
    evidence = prior * p_evidence_if_true + (1 - prior) * p_evidence_if_false
    return prior * p_evidence_if_true / evidence

print(round(bayes(0.01, 0.95, 0.10), 3))   # 0.088
```

Check by simulation

```
import numpy as np
rng = np.random.default_rng(3)
people = 1_000_000
sick = rng.random(people) < 0.01
positive = np.where(sick, rng.random(people) < 0.95, rng.random(people) < 0.10)
print(round(sick[positive].mean(), 3))   # close to 0.088
```

Updating again

A second, independent positive test? Use the first posterior as the new prior:

```
p1 = bayes(0.01, 0.95, 0.10)
p2 = bayes(p1, 0.95, 0.10)
print(round(p1, 3), round(p2, 3))   # 0.088 -> about 0.48
```

Each piece of evidence moves the belief. This is how a careful doctor — and a careful AI — reasons.

The base-rate mistake

Ignoring how common something is (the **base rate**, or prior) leads to wrong conclusions: *"the fraud system flagged this customer, so they must be a fraudster"*. If fraud is rare, most flags may be false alarms. Always ask: **how common was this before the evidence?**

A spam filter uses Bayes

A **Naive Bayes** spam filter learns, from labelled emails, how often words appear in spam and in normal mail:

- P(spam) = 0.4 (40% of emails are spam)
- P("winner" | spam) = 0.20, P("winner" | normal) = 0.01

```
print(round(bayes(0.4, 0.20, 0.01), 3))   # 0.93: an email with "winner" is probably spam
```

It combines the evidence from many words to give each email a probability. Simple, fast and still used today.

Key terms

- **Prior** — the probability before seeing new evidence.
- **Likelihood** — how probable the evidence is if the belief is true.
- **Posterior** — the updated probability after the evidence.
- **Base rate** — how common something is in the population.
- **Naive Bayes** — a classifier that applies Bayes' theorem to many features.

Remember this: Bayes' theorem updates a belief with evidence: posterior ∝ likelihood × prior. When the prior is small, even a good test produces many false alarms — always consider the base rate.

Name: Date:

Activity

Task 1

Recalculate the disease example if the disease affects 10% of people. What changes, and why?

Task 2

Improve the test: false positives drop from 10% to 1%. What is P(sick | positive) now?

Task 3

The word "free" appears in 25% of spam and 5% of normal mail; P(spam) = 0.4. What is P(spam | "free")?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. A disease affects 1% of people; the test catches 95% of sick people and has 10% false positives. P(sick | positive) is about…
   A. 50%
   B. 95%
   C. 9%
   D. 1%
2. In Bayes' theorem, what is the prior?
   A. The probability before seeing the new evidence
   B. The test's accuracy
   C. The probability after the evidence
   D. The number of people

3. What is the posterior?

A. The false positive rate
B. The updated probability after the evidence
C. The base rate
D. A chart

4. Why do rare conditions produce many false alarms even with good tests?

A. The healthy group is so large that its small error rate creates many false positives
B. Tests are always bad
C. It never happens
D. Computers are slow

5. After one positive test, P(sick) became 0.088. For a second independent positive test, what do you use as the new prior?

A. 0.95
B. 0.088
C. 0.10
D. 0.01

6. Which AI tool is built directly on Bayes' theorem?

A. k-means
B. A CSV file
C. Naive Bayes spam filter
D. Matplotlib

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